

PHYS 3018 BSc Topics 2025/26

Dear Students,

This note is for all 2nd year students that they are registered for a three year programme and will get the PHYS3018 (BSc projects) module next year. In case any of you have a last minute change to a 4th year program, please do not submit BSc project preferences and inform me.

I have attached a booklet containing titles and abstracts for the BSc projects we anticipate will be on offer for the coming academic year (25/26). First, please read carefully the projects and then visit the link

<https://www.licn.phys.soton.ac.uk/scriptBSC/index.php>

and submit your top 6 choices of projects.

We will endeavour to allocate you to one of your choices. Apart from the BSc projects advertised in the booklet, you are strongly encouraged to contact any supervisor with your own idea for a project. In case that you reach an agreement for a new project, you will first need to find a project partner and then please contact the module coordinator to discuss the allocation.

Please try to keep a balance of choices between the different thematic domains. The deadline to submit your preferences is **Friday 2nd May 2025 at midnight**. Late submission of this form will almost guarantee that your choices are not met, so please submit on time.

ASTRONOMY GROUP

BA01. Supervisor: Dr. Daniel Whiter

Topic: Space Environment Physics

Title: Pulsations in the aurora

The aurora (a.k.a. northern lights) is caused by energetic particles precipitating into the atmosphere and ionising and exciting oxygen and nitrogen molecules and atoms, which emit light of different wavelengths. Pulsations in the aurora are observed as on-off modulation of the auroral brightness with periods of a few seconds to a few 10s of seconds. They are thought to be caused by wave-particle interactions in space, in the equatorial plane of Earth's magnetosphere. Pulsating aurora has received quite a lot of interest in recent years because it includes very energetic electron precipitation which can affect the chemistry and climate of the middle and upper atmosphere, for example contributing to production of nitric oxide and ozone destruction. Pulsating aurora is typically seen in the early morning hours, equatorward of the main auroral oval, for example over Lapland. Occasionally they are seen as far north as

Svalbard, in the high-Arctic, although so far there have been very few studies of pulsating aurora at such high latitudes. In this project you will use both ground-based and spacecraft data to investigate the geomagnetic and solar wind conditions during a set of pulsating aurora events seen from Svalbard, with the aim of identifying the conditions required for the pulsations to reach so far north.

BA2. Supervisor: Prof. Ian McHardy

Topic: Astronomy

Title: Analysis of Stock Market Variability (up to 4 students)

Almost everything in the universe, including the price of stocks and shares, varies. But how can we quantify and characterise the variability? Is there any deep driving force underlying the variability of the stock market or is it purely a random noise process? In this popular project students will apply some of the techniques which we currently apply to the analysis of the X-ray and optical variability of stars and galaxies to try to make sense of variations in stock market price and to determine whether there is any predictability in the stock market, ie can we make any money out of it! The techniques which we will use include 'cross correlation' and 'power spectral' analysis. Students will write simple python programmes for these analyses. With cross-correlation functions we will determine whether there is any relationship between the variability in two different stock market prices and whether the variations in one stock might lag behind the variations in the other. With power spectral analysis we will search for any characteristic timescales in the data, eg periodicities. Knowledge of these highly transferable skills has proven very valuable to students seeking jobs in the financial and tech sectors.

BA3. Supervisor: Prof. Ian McHardy

Topic: Astronomy

Title: Mapping the Inner Structures of Active Galactic Nuclei (AGN)

The inner geometry of AGN is not at all well known. These nuclei are far too small to image directly but we can use a technique called 'Reverberation Mapping'. In this technique X-ray emission from around the central supermassive black hole (SMBH) shines on the surrounding material, principally the accretion disc, and makes it reverberate, i.e. the X-rays are reprocessed and boost the existing emission from that material. As that material is cooler than the X-ray source, the reprocessed emission is UV or optical, depending on the temperature. By measuring the lag between the direct X-rays and the reprocessed UV/optical emission, we can determine how far apart those emission regions are. We can measure lags for lots of different UV and optical bands and hence map out the inner structures of AGN. In the first part of this project the students will write a simple python cross-correlation programme to measure lags from an exciting new set of AGN X-ray/UV/optical lightcurves. In the second part the students will compare their observed lags with model lags which they will calculate, based on the mass and accretion rate of the AGN. Hence they will test whether our previously accepted model of an accretion disc is correct. If it is not completely correct, students might find reasons as to why not! This project has some technical similarities to BA2 but is aimed more at students interested in astronomy. The technical skills learned here are again very valuable for those seeking jobs in the tech sector.

BA4. Supervisor: Prof. Francesco Shankar

Topic: Astronomy

Title: Galaxy Clustering

The spatial distribution of galaxies is usually referred to in the literature as "Clustering". Systematic measurements of galaxy clustering can allow us to infer the type of dark matter haloes which should contain the galaxies of a given type and luminosity. The aim of this project is to predict the clustering of a subset of galaxies via the use of large-scale dark matter simulations. Galaxies will be assigned to host dark matter haloes via a variety of semi-empirical relations which will then be tested by proper comparison with a variety of independent observational data. Information on galaxy evolution will then be extracted by comparing the stellar-to-halo mapping at different redshifts.

BA5. Supervisor: Dr. D. Altamirano

Topic: Astronomy

Maximum students: 6

Title: Using Astronomy-related techniques to tackle world problems

The Global Challenges Research Fund (GCRF) is a 5-year £1.5Bn fund and a key component in the delivery of the UK Aid Strategy: tackling global challenges in the national interest. During this project, the student will use all the knowledge acquired so far at the University to investigate possible solutions for problems within the GCRF framework. This project will involve identifying the problem and their possible solutions (e.g. using data analysis, complex algorithms, new technologies, IoT, or just common sense). Past examples include: (a) designing an early warning system for Malaria in Children living in Sub-saharan Africa (which covered, among others, calculating probability of infection, modeling systematic errors in the measurement of the body temperature, calculating the efficiency of micro-solar panels, etc); (b) Identify unsustainable cities based on the satellite image analysis which reveals the ratio between green and building space, and a statistical study of how that ratio changes the quality of the air we breathe; (c) Understanding the financial stability of countries based on the time-series analysis of their GDP evolution in time; (d) Studying the benefits of introducing renewable wind energy in developing countries (which covered the critical study of wind-energy efficiency vs the current costs of alternative methods, modelling the number of wind-mills needed to feed a city with electricity, etc).

BA6. Supervisor: Prof. Christian Knigge

Topic: Astronomy

Title: The Mass-Radius Relation of Donor Stars in Accreting Compact Binaries

Accreting compact binaries are systems in which a white dwarf, neutron star or black hole gains mass from an accretion disk that is fed by a mass-losing low-mass companion star. This donor star is almost a normal main-sequence star -- but not quite. Because of the mass loss it experiences, it is always slightly out of equilibrium. As a result, its radius is a little bit larger than that of a normal main-sequence star with the same mass. This radius inflation is a great way to measure the mass-loss rate, since the faster mass is lost, the more the star will be inflated. The mass-radius relation of these donor stars can therefore be used to track the long-term evolution of these binary systems. In this project, we will collect data sets for donor stars from

the astronomical literature and use these to construct mass-radius relations for compact binaries with white dwarf, neutron star and black hole primaries. We will then compare these relations to theoretical predictions.

BA7. Supervisor: Dr. Imogen Gingell

Topic: Space Environment Physics

Title: Trapping Particles in Magnetic Structures

Planetary magnetospheres are highly dynamic environments, influenced by the inflowing solar wind at all spatial scales. At “kinetic” scales, single particle trajectories strongly influence the generation and evolution of plasma instabilities. For example, the shear flow between the solar wind and magnetospheric plasmas can create “Kelvin-Helmholtz” vortices in the flanks of planetary magnetospheres. At Mercury, metal ions can become trapped within these vortices, and influence the scales at which they grow (Gingell et al. 2015). Closer to home at Earth’s bow shock, small ripples can act as sites of local ion acceleration (Sundberg et al. 2016). The aim of this project is to determine under what conditions ions or electrons can become trapped or accelerated by these kinds of structures using “test particle” simulations. This project will involve programming (using e.g. Python/C++/Matlab) to solve particle trajectories and analyse simulation data, so proficiency with computing will be an advantage.

BA8. Supervisor: Dr. Robert Fear

Topic: Space Environment Physics

Title: How quickly does the Earth’s magnetosphere respond to the solar wind?

The Earth’s magnetic field forms a cavity in the solar wind called the magnetosphere; the interaction between the solar wind and the magnetosphere is ultimately responsible for dynamics of near-Earth space, including variations in the intensity of the radiation belts and the most spectacular displays of the aurora (the northern and southern lights), and is the ultimate cause of the most severe ‘space weather’ events. The night side of the Earth’s magnetosphere forms an extended magnetotail. It has long been known that the east-west (B_y) component of the magnetic field observed in the magnetotail is proportional to the equivalent component observed in the solar wind, but the timescales on which this happens are disputed. In this project, you will use satellite data from the European Space Agency’s Cluster mission to investigate the timescales on which the magnetotail is controlled by the interplanetary magnetic field. Basic computer programming ability will be an advantage for this project.

BA9. Supervisor: Dr Robert Jeffrey

Topic: Astronomy

Title: Exploring accreting compact objects through time series analysis

Some of the most dynamic astronomical objects in the Galaxy are accreting compact objects in binary systems. As they are both highly energetic and small in size, these systems can vary on short, observable timescales. There are now several monitoring campaigns (particularly from the MAXI, NICER and Swift x-ray instruments) which have produced multi-year, high cadence data from many of these sources. In this project, we will compare different ways to analyse these time series to infer properties of these systems, working in both the signal and Fourier domains. You will use techniques for time series analysis that have applications across

astronomy, experimental physics, computing, engineering, finance, and data science. Some familiarity with data handling in Python (or a similar language) will be important for this project.

THEORY GROUP

BT1. Supervisor: Prof. Alexander Belyaev (up to 4 students)

Topic: Theoretical and Computational Physics

Title: “Dark Matter and Beyond: Uncovering the Mysteries of the Universe”

The nature of Dark Matter (DM) is one of the most pressing questions in modern particle physics. Its existence, inferred from multiple cosmological experiments, reveals that DM comprises about 95% of the Universe's energy content, with ordinary matter contributing just 5%. The discovery of DM in Earth-based experiments would be more groundbreaking than the 2012 Higgs boson discovery, as it would confirm physics beyond the Standard Model (SM). This project is dedicated to exploring the most promising DM models, and it requires a deep understanding of both theoretical particle physics and advanced computing techniques. Students will focus on:

1. Mastering the Standard Model (SM), its particles, and their interactions.
2. Investigating simple yet fundamental extensions of the SM that include DM particles with various spins.
3. Exploring complex processes leading to distinctive DM signatures.
4. Assessing the potential of cutting-edge experiments to identify and characterize DM particles.

This project is designed for highly motivated students with an exceptional background in theoretical physics and advanced programming skills. If you are passionate about making a significant contribution to one of the biggest challenges in particle physics, this opportunity is for you. Strong problem-solving abilities and proficiency in computational techniques will be essential.

BT2. Supervisor: Prof. S Moretti

Topic: Theory

Title: *The quantum anharmonic oscillator*

While the solution to the Schrodinger equation describing a simple harmonic oscillator is a textbook example, the case of an anharmonic oscillator is far more challenging. The aim of this project is to review the solutions attempted in specialised literature for a generic anharmonic oscillator, by discussing the techniques adopted and distinguishing between the cases where analytical solutions have been found and where only numerical ones are available. Depending on time and progress, the project may involve the use of worked out computer examples for the latter case. A very good understanding of the Schrodinger equation, ability in solving differential as well as integral equations and strong generic algebraic and/or numerical skills are required.

BT3. Supervisor: Prof. N. Evans

Topic: Theory

Title: Relativistic Quantum Mechanics, Renormalization & Theoretical Constraints on the Higgs Boson

The Higgs boson's discovery at the LHC particle collider is the discovery of a new type of matter– it is a fundamental, relativistic scalar particle. We will explore the Klein Gordon relativistic wave equation for such scalars, compute the effects of interactions in perturbation theory and see how infinities occur beyond leading order. These can be renormalized away but lead to some important constraints on the validity of the Higgs theory which we will explore.

BT4. Supervisor: Prof. N. Evans

Topic: Theory

Title: Quark Interactions From AdS-Space

String theory originated from the study of the spectrum of mesons. Its successes though were incorporated into the quark model in which the mesons are a quark and an anti-quark tied by chromoelectric flux. Indeed string theory floundered since it needs to live in ten dimensions. The link to mesons seemed lost until a recent realization that the string between the quarks can be used to model the quarks interactions, the string just lives in a higher dimension curved spacetime! In this project we will construct the Nambu-Goto string action then solve Euler Lagrange equations for quark anti-quark configurations. Electromagnetic like inter-quark potentials can emerge from Anti-de-Sitter space, confinement from hard wall geometries and temperature can be included via black holes.

BT5: Supervisor: Prof P. Di Bari

Topic: Theory

Title: Thermal history of the early universe

Understanding how the temperature evolved in the early universe is a nice application of thermodynamics and statistical mechanics to the early universe and the temperature of relic neutrinos would be a key parameter. In the project the calculation of the ultrarelativistic degrees of freedom of the primordial plasma will be done applying general statistical mechanics, as studied in the PHYS2024 module, extended to the relativistic regime. As additional ingredient a new particle species can be added to the standard lore, for example a new particles species that might be related to the dark matter of the universe. In this case the change in the prediction of the relic neutrino temperature will be an interesting result for new physics search in cosmology and that might be related to the current Hubble tension in cosmology. (NOTE: Attending PHYS6005 Cosmology is not a pre-requisite, though it might help, basics of cosmology will be given in initial lectures. The project will mainly require knowledge of Statistical Mechanics at the level of PHYS2024)

BT6. Supervisor: Dr Bipasha Chakraborty

Topic: Theory/computation

Title: Machine learning for lattice Quantum Chromodynamics

Quantum Chromodynamics is the theory of the Strong force, and it encodes how quarks and gluons interact with each other and form most of the visible matter in the Universe. Because of

the nature of the Strong force, it cannot be analytically solved at low energies, and is only viable to first-principles computational techniques known as lattice QCD. Lattice QCD requires a huge amount of computational resources and produces a huge amount of data to analyse. The aim of this project is to explore various machine learning techniques for the analysis of lattice QCD data. [Please note, this does not require previous knowledge of lattice QCD or machine learning].

BT7. Supervisor: Dr Debasish Banerjee

Topic: Theory/computation

Title: Exact Diagonalization and Tensor Networks for Spins and Fermions

Outline: Strongly interacting systems have the microscopic degrees of freedom interacting with each other very strongly. They often give rise to phases which are exotic, contain fractionalized excitations and fundamentally different from the original degrees of freedom themselves. In this project, we will learn two tools which can be reliably used to study such systems --- Exact Diagonalization (ED) and the tensor network approach. While the former can obtain all details of the system, the latter consists of a systematic approximation to learn certain details of the system.

BT8. Supervisor: Dr Bipasha Chakraborty

Topic: Theory/computation

Title: Quantum computing for fundamental physics

Quantum computing has emerged as a promising tool for solving complex problems in fundamental physics, particularly in areas where classical computational approaches face limitations. This project explores the application of quantum algorithms to key problems in fundamental physics. Specifically, we investigate how quantum circuits can simulate fundamental interactions and solve quantum many-body problems.

QLM GROUP

BQ01. Supervisor: Prof. M. Kaczmarek

Topic: QLM

Title: Liquid crystals for vision correction

Presbyopia, an age-related decline in near vision, or myopia are common conditions that require correction either with glasses or contact lenses. However, achieving clear vision at various distances typically involves using multiple pairs of glasses or varifocal lenses, which come with some limitations. There is significant interest in developing lenses that can dynamically adapt their focal length depending on the viewing distance and liquid crystals are the main candidates for this technology. Liquid crystals have semi-crystalline order as well as fluid nature and have effective birefringence that can be controlled by electric field. When used as an active material in lenses, the focal length can be adjusted with electric field, which can be controlled by the circuit, incorporated in the frame, for example.

In this experimental project, you will assess different liquid crystals for their potential use in lens applications. Given the wide variety of liquid crystal materials, the first step will involve identifying those with physical and optical parameters that are most promising for lenses. The experiments will then investigate selected liquid crystals, examining the effect of different thicknesses of liquid crystal layers and varying electric field, on light transmission.

BQ2 Supervisor Prof. D.C. Smith

Topic: QLM

Title: Build Your Own Optical Microscope

The aim of this project will be to build from components an optical microscope and test its capabilities. Whilst this might sound easy, I've done this myself and it turns out to be much more challenging than you might imagine. If you achieve a reasonable microscope then the next stage will be to investigate how diffraction affects the image produced. Some practical photonics experience will make this project considerably easier.

BQ3. Supervisor: Professor Hendrik Ulbricht

Topic: QLM

Title: Single particle heat engine using levitated optomechanics

This experimental project is about the manipulation of the centre of mass motion of a single trapped glass particle in a vacuum chamber. To this end you will use a laser to trap the particle while modulating the intensity and frequency of the laser. The single laser beam is used for trapping the particle, detection particle's motion [position and momentum] and manipulating particle's motion [cooling, squeezing, heating]. The aim of this project is to realise a manipulation scheme, which is equivalent to a thermodynamical cycle, such as the Otto-cycle or the Stirling-cycle.

BQ4. Supervisor: Professor Hendrik Ulbricht

Topic: QLM

Title: Detection of black-body emission of a single nanoparticle in vacuum

This experimental project is about the spectroscopy of the emitted light from a single particle in an optical trap. In levitated optomechanics experiments we study the centre of mass motion of a single particle trapped by optical forces. This can be used for instance to cool the motion to low temperatures and to the quantum mechanical ground state of the harmonic motion. However, the internal temperature of the particle, while at very low motional temperature, is uncontrolled and in many cases even unknown. In this project you will work to measure the emitted black-body spectrum of the trapped particle and use it to determine the particle's internal temperature.

BQ5. Supervisor: Dr. Tim Freegarde

Topic: QLM

Title: Absolute electronic measurement of the speed of light

According to Maxwell's well-tested theories, the resonant frequency of a simple L-C circuit depends only upon the component dimensions and the speed of light. By constructing from scratch a range of air-spaced capacitors and inductors, measuring the resonant frequencies of

their combinations, and carefully accounting for measurement and end effects, this project allows the speed of light to be determined absolutely in a benchtop experiment.

BQ6. Supervisor: Dr. Marcus C. Newton

Topic: QLM

Title: Deep Learning Phase Retrieval

Phase retrieval is the process of recovering images and phase information from Fourier amplitude measurements of a diffraction pattern. There are several approaches to recovery each of which generally require the use of computational algorithms. Deep learning is a branch of artificial intelligence (AI) that has emerged as a novel and powerful approach to phase retrieval. Deep learning employs a convolution neural network (CNN) or a generative adversarial network (GAN) to solve the phase retrieval problem. In this project you will learn how to design, train and optimise a convolution neural network to solve the phase retrieval problem for diffraction patterns obtained from coherent x-ray diffraction imaging experiments.

BQ7. Supervisor: Prof. O. Muskens

Topic: QLM

Title: Testing lidar for applications in road safety

Applications of lidar are seeing a tremendous rise in recent years, in particular in relation to autonomous vehicles. Next to highly sophisticated 360 degree spinning lidar domes that cost more than the price of a new car, lower cost and compact solid-state systems are of great interest for a variety of applications. This requires some trade-offs in how signals are generated, collected, and processed. The brute-force approach of collecting 3D point clouds of millions of samples per second requires significant computational overhead and more efficient methods may be needed. In this project the student will investigate using off the shelf lidar units, test their performance and explore different ways of looking at the data. Some experience with programming is desirable.

BQ8. Supervisor: Dr Silvia Motti

Topic: QLM

Title: Semiconductor charge dynamics and photoluminescence

Photoactive semiconductors are a key component of many important devices, such as solar cells, LEDs, lasers, and colour displays. This project will explore the key properties of semiconductor materials that are relevant for technological applications. You will use numerical methods to model the dynamics of charges in the semiconductor and model the mechanisms of photon emission. some experience programming in Python/Matlab is desirable.

BQ9. Supervisor: Bruce (Jun-Yu) Ou

Topic: QLM

Title: Imaging with Flat optics

Flat optics(metalens) made by artificial nanostructures has powerful ability of controlling amplitude, phase, and polarization of light on demand with nanometre precision. Metalens can be used to build compact and multifunctional microscopes with small volume and light weight

compared to conventional, bulky, and single function optical elements. For example a visible compact zoom microscope with three magnifications constructed by two pieces of glasses (metalenses) with on-axis tri-focal planes and no moving parts. In this project, you will understand the principle of an imaging system, measure the wavefronts, and the polarisation dependences of metalenses fabricated in Southampton cleanroom complex. Some familiarity with programming/Matlab is useful for the project.